

TOI-1811b

R-2106

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_260302 · created 2026-08-10T23:50:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461101.7198 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1642 +/- 0.0091
Transit depth (Rp/Rs)^2	2.7 +/- 0.3 [%]
Orbital Inclination (inc)	87.37 +/- 0.87
Airmass coefficient 1 (a1)	1.14 +/- 0.017
Airmass coefficient 2 (a2)	-0.104 +/- 0.0071
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	None
Optimal Aperture	3.93
Optimal Annulus	7.47
Transit Duration (day)	0.093 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1642 ± 0.0091 vs 0.13272 (deviation 3.26σ)
Duration fit vs published	0.093 d vs 0.0813 d (deviation 1.15σ)
Mid-time O–C	-0.3 min
Depth SNR	17.0
Residual scatter	1.45 %

Light curve

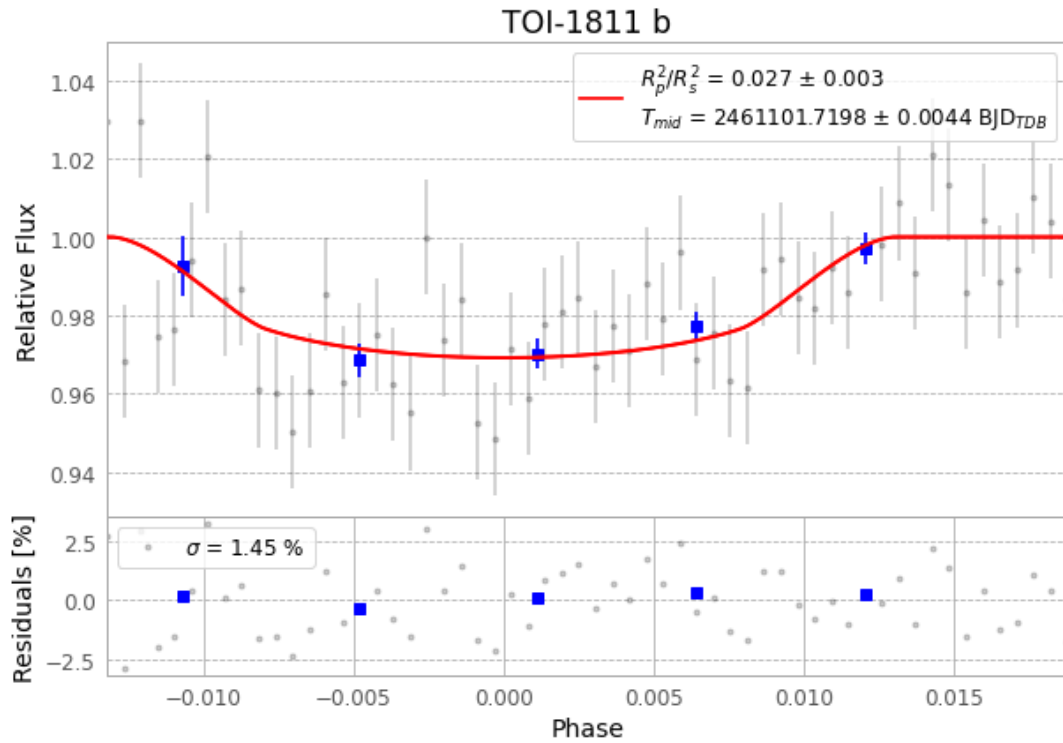


Figure 1 — FinalLightCurve_TOI-1811 b_2026-03-01.png (SHA-256 268344845eb1a1eb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [229, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 92db689e792998eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), TOI1811260302035716.FITS → TOI1811260302064817.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-260302.log (4f2cbf2a9476...), FinalLightCurve_TOI-1811 b_2026-03-01.png (268344845eb1...), FinalLightCurve_TOI-1811 b_2026-03-01.csv (cb9c0bcefad3...)

Compute resources

Wall time 141.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:50Z.